Worksheet 6

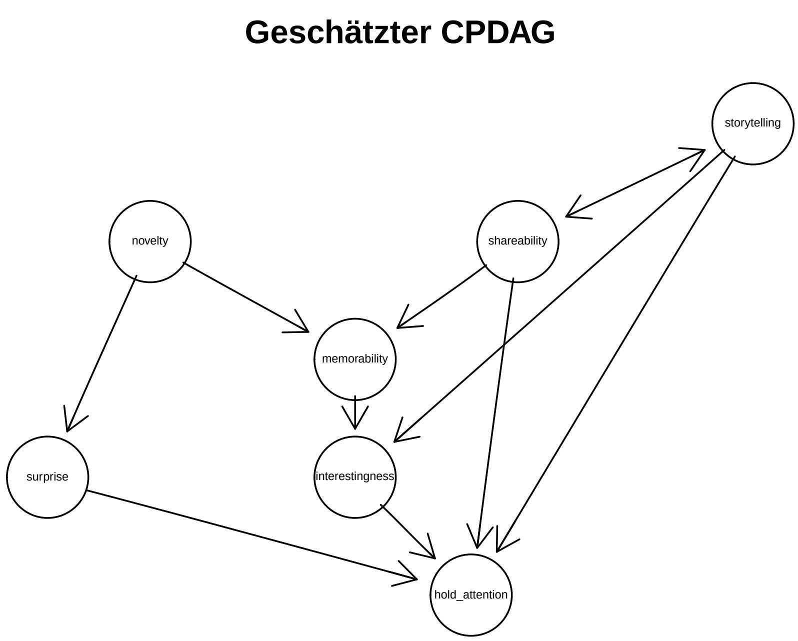
Exercise 1: Webcams

In a research and art project titled "Watching the World" (https://webcamaze.engineering.zhaw.ch/), AI is used to extract interesting images from publicly accessible webcams. In a study, participants were asked about various aspects of randomly selected webcam images. The goal is to determine what makes an image interesting. The study data is available in the file `interesting.rda`.

```
load("data/interesting.rda")
```

a) Estimate the causal structure using the PC algorithm. Visualize the estimated CPDAG. Use alpha = 0.0001.

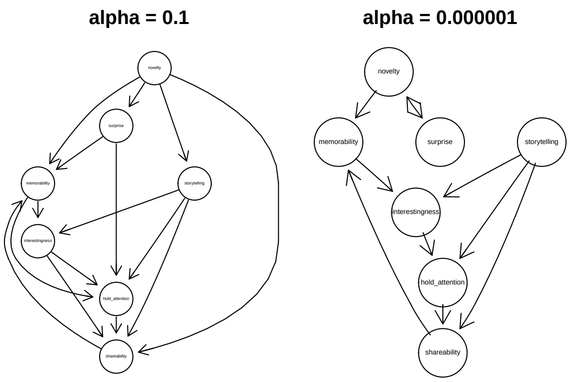
```
library(pcalg)
suffStat <- list(C = cor(d.interesting), n = nrow(d.interesting))
pc.fit <- pc(suffStat, gaussCItest, alpha = 0.0001,
                labels = colnames(d.interesting))
plot(pc.fit, main = "Geschätzter CPDAG")
```



b) Test the sensitivity of the PC algorithm to different α values.

```
fit.pc.1 <- pc(suffStat,
           indepTest = gaussCItest,
           labels = colnames(d.interesting),
           alpha = 0.1)
fit.pc.2 <- pc(suffStat,
           indepTest = gaussCItest,
```

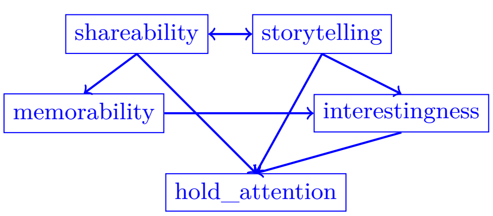
```
        labels = colnames(d.interesting),
        alpha = 0.000001)
par(mfrow = c(1,2))
plot(fit.pc.1, main = "alpha = 0.1")
plot(fit.pc.2, main = "alpha = 0.000001")
```



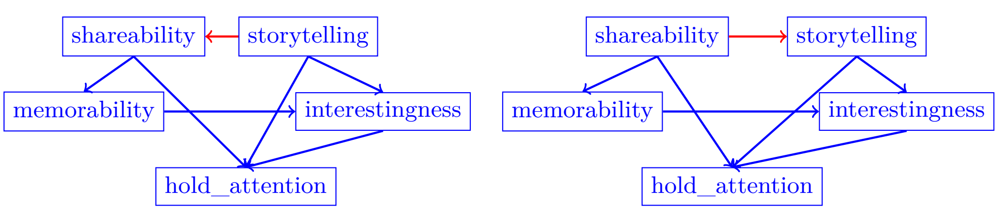
In general, a smaller α results in a graph with less edges, while a larger α leads to a more dense graph. However, we must consider the issue of multiple testing:
For $\alpha = 0.1$ there are 502 tests of conditional independence (42 tests with conditioning on no variable, 185 tests conditioning on one variable, 188 conditioning on two variables, 72 conditioned on 3 and 15 conditioned on 4). A simple Bonferroni correction suggests an α which is about 0.1/502 = 0.0002 for an over-all significance of 0.1. This is probably too pessimistic. A choice of $\alpha = 0.1$ is definitely too weak.

c) Have a look at the estimated subgraph for the 5 variables: `storytelling`, `shareabilty`, `hold_attention`, `memorability` and `interestingness`. Report all DAGs that belong to the Markov equivalence class of the estimated CPDAG subgraph.

Estimated CPDAG subgraph:



Markov equivalence class:



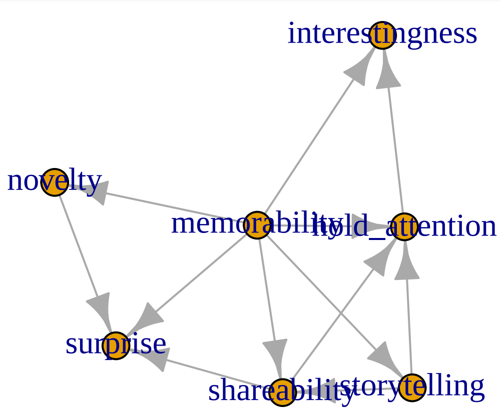
d) Estimate the causal structure using the LiNGAM algorithm under the assumption of non-Gaussian errors. Visualize the estimated DAG.

```
# LinGAM:
fit <- lingam(d.interesting)
colnames(fit$Bpruned) <- rownames(fit$Bpruned) <- colnames(d.interesting)
round(fit$Bpruned,2)
```

```
                novelty surprise storytelling memorability shareability hold_attention
novelty            0.00        0         0.00         0.45         0.00           0.00
surprise           0.67        0         0.00         0.33        -0.29           0.00
storytelling       0.00        0         0.00         0.49         0.00           0.00
memorability       0.00        0         0.00         0.00         0.00           0.00
shareability       0.00        0         0.47         0.69         0.00           0.00
hold_attention     0.00        0         0.21         0.33         0.66           0.00
interestingness    0.00        0         0.00         0.32         0.00           0.47
                interestingness
novelty                       0
surprise                      0
storytelling                  0
memorability                  0
shareability                  0
hold_attention                0
interestingness               0
```

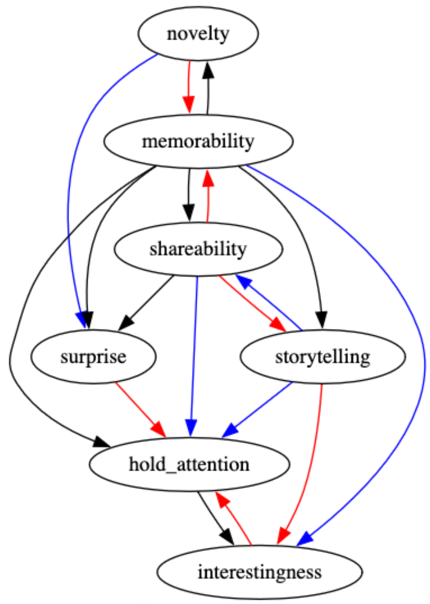
```
# Estimated causal graph
library("gRbase")
adj_matrix <- t(ifelse(fit$Bpruned != 0, 1,0))
mx <- as(adj_matrix,"igraph")
```

```
plot(mx)
```



e) Compare the two estimated graphs from (a) and (b).

A comparison of the results from the PC algorithm and LiNGAM shows 5 edges (in blue) that are identical in both graphs. There are 6 edges (in red) that were found only by the PC algorithm and 7 edges (in black) that were detected only by LiNGAM.



The algorithms for learning causal structures produce very different graphs. They also make very different assumptions about the data: the PC algorithm assumes a normal distribution, while LiNGAM does not. The results of these algorithms should be reviewed by subject matter experts to ensure their plausibility and should not be used directly as truecausal graphs for further analysis. Additionally, the stability of both the LiNGAM and PC algorithms could be assessed by subsampling the data. This allows to identify stable edges that appear in at least 90% of 100 subsampling runs.

d) Based on the results of the LiNGAM algorithm, estimate the following total causal effects:
   - of `novelty` on `surprise`
   - of `storytelling` on `interestingness`
   - of `memorability` on `interestingness`
   - of `suprise` on `interestingness`

There is a direct effect from `novelty` to `surprise`:

```
fit$Bpruned["surprise","novelty"]
```

```
[1] 0.6715825
```

The causal effect from `storytelling` to `suprise` arises via the path through `shareability`:

```
c1 <- fit$Bpruned["shareability","storytelling"]
c2 <- fit$Bpruned["surprise","shareability"]
c1 * c2
```

```
[1] -0.1375445
```

The causal effect from `memorability` to `interestingness` is based on 4 causal pathways.

```
# Path 1: memorability -> shareability -> hold_attention -> interestingness
p1 <- fit$Bpruned["shareability","memorability"] *
      fit$Bpruned["hold_attention","shareability"] *
      fit$Bpruned["interestingness","hold_attention"]
# Path 2: memorability -> hold_attention -> interestingness
p2 <- fit$Bpruned["hold_attention","memorability"] *
      fit$Bpruned["interestingness","hold_attention"]
```

```
# Path 3: memorability -> storytelling -> hold_attention -> interestingness
p3 <- fit$Bpruned["storytelling","memorability"] *
      fit$Bpruned["hold_attention","storytelling"] *
      fit$Bpruned["interestingness","hold_attention"]
# Path 4: memorability -> interestingness
p4 <- fit$Bpruned["interestingness","memorability"]
# Total Effect:
p1 + p2 + p3 + p4
```

```
[1] 0.7374333
```

There is no causal path from `surprise` to `interestingness`, so the causal effect is 0.

Exercise 2: Travel Time

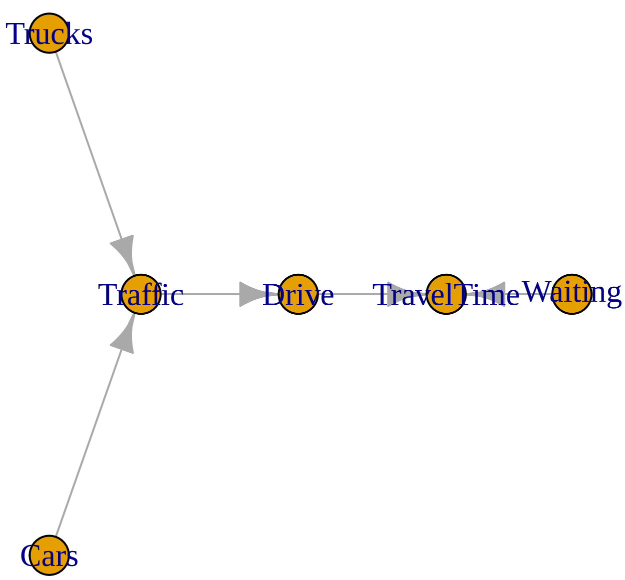
The dataset `bus.rda` contains simulated travel times for a long-distance bus journey from Zurich to Rome.

```
load("bus.rda")
```

a) Estimate the causal structure using the LiNGAM algorithm, assuming Non-Gaussian noise. Visualize the estimated DAG.

```
# Estimating the causal structure with LiNGAM
fit.bus <- lingam(bus)
# Drawing the estimated DAG
library("gRbase")
adj_matrix <- t(ifelse(fit.bus$Bpruned != 0, 1,0))
colnames(adj_matrix) <- rownames(adj_matrix) <- colnames(bus)
mx <- as(adj_matrix,"igraph")
```

```
plot(mx)
```



b) Based on the DAG estimated in (a), which variables are direct causes of travel time?

The estimated DAG shows two incoming arrows to travel time: Waiting and Drive.

c) What is the total causal effect of cars on travel time according to estimated LiNGAM model?

```
B <- round(fit.bus$Bpruned,2)
colnames(B) <- rownames(B) <- colnames(bus)
B
```

```
          Waiting Traffic Trucks Cars Drive TravelTime
Waiting      0.00    0.00   0.00    0  0.00          0
Traffic      0.00    0.00   1.97    1  0.00          0
Trucks       0.00    0.00   0.00    0  0.00          0
Cars         0.00    0.00   0.00    0  0.00          0
Drive        0.00    4.03   0.00    0  0.00          0
TravelTime   0.99    0.00   0.00    0  0.94          0
```

The total causal effect is the sum over all directed paths from Cars to Travel Time. There is only one directed path: Cars $\rightarrow$ Traffic $\rightarrow$ Drive $\rightarrow$ Travel Time. The causal effect of a directed path corresponds to the multiplication of the edge weights along the directed path. The estimated edge weights can be read off from the B matrix, i.e. $1 \cdot 4.03 \cdot 0.94 = 3.79$.

d) How does the estimated linear SCM with non Gaussian noise look like?

```
# coefficients
B
```

```
          Waiting Traffic Trucks Cars Drive TravelTime
Waiting      0.00    0.00   0.00    0  0.00          0
Traffic      0.00    0.00   1.97    1  0.00          0
Trucks       0.00    0.00   0.00    0  0.00          0
Cars         0.00    0.00   0.00    0  0.00          0
Drive        0.00    4.03   0.00    0  0.00          0
TravelTime   0.99    0.00   0.00    0  0.94          0
```

```
# intercepts
cx <- fit.bus$ci
names(cx) <- colnames(bus)
round(cx,2)
```

```
   Waiting    Traffic     Trucks       Cars      Drive TravelTime
     19.91       6.21      30.62      84.64      34.47      17.61
```

Estimated linear SCM by the LiNGAM algorithm:
Waiting $\leftarrow 19.91 + E_W$
Trucks $\leftarrow 30.62 + E_{Tr}$
Cars $\leftarrow 84.64 + E_C$
Traffic $\leftarrow 6.21 + 1.97 \cdot$ Trucks $+$ Cars $+ E_T$
Drive $\leftarrow 34.47 + 4.03 \cdot$ Traffic $+ E_D$
TravelTime $\leftarrow 17.61 + 0.99 \cdot$ Waiting $+ 0.94 \cdot$ Drive $+ E_{TT}$

e) Assume Marc traveled today from Zurich to Rome. He had to wait 40 minutes for the long-distance bus and his total travel time was 742 minutes. The actual driving time was 640 minutes and the traffic volume was 133. What would his travel time have been if the traffic volume would had only been 110?

Observed data:

Waiting = 40, TravelTime = 742, Drive = 640, Traffic = 133
Calculating counterfacutal:
1. Abduction

```
(Ett <- 742 - (17.61 + 0.99 * 40 + 0.94 * 640))
```

```
[1] 83.19
```

```
(Ed <- 640 - (34.47 + 4.03 * 133))
```

```
[1] 69.54
```

```
(Ew <- 40 - 19.91)
```

```
[1] 20.09
```

2. Action: Modified SCM for do(Traffic = 110):
Waiting $\leftarrow 19.91 + E_W$
Traffic $\leftarrow 110$
Drive $\leftarrow 34.47 + 4.03 \cdot$ Traffic $+ E_D$
TravelTime $\leftarrow 17.61 + 0.99 \cdot$ Waiting $+ 0.94 \cdot$ Drive $+ E_{TT}$

3. Prediction

```
Waiting <- 19.91 + Ew
Traffic <- 110
Drive <- 34.47 + 4.03 * Traffic + Ed
(TravelTime <- 17.61 + 0.99 * Waiting + 0.94 * Drive + Ett)
```

```
[1] 654.8714
```

If the traffic volume would have been only 110, Marc's travel time would have been 742 - 655 = 87 minutes shorter.